UNIVERSITY OF DERBY
College of Science and Engineering

Mock Paper 2324

OPERATIONAL RESEARCH
5CM522

DATE: REVISION MATERIAL
TIME ALLOWED: 4 HOURS

Instructions to Candidates

Answer all 4 questions.

This is a reformatted mock paper designed to support preparation for the timed in-class test.

You may use Excel or Python, but your workings, files and interpretations must be clear and well labelled.

This version keeps the topics most relevant to the current module revision focus.

The original EOQ / inventory question from the older mock has been omitted because it is not part of the current test coverage.

1.

The data analyst at N&F has collected the following seasonal sales data.

Period	Season	Sales
1	Spring	1013
2	Summer	1025
3	Autumn	1002
4	Winter	1021
5	Spring	1053
6	Summer	1067
7	Autumn	1042
8	Winter	1060
9	Spring	1093
10	Summer	1106
11	Autumn	1083
12	Winter	1101
13	Spring	1134
14	Summer	1147
15	Autumn	1123
16	Winter	1139

Use periods 1–12 as the fitting sample and periods 13–16 as a validation set.

(a) Calculate suitable centred 4-period moving averages for periods 1–12 and estimate the trend where possible. Plot or tabulate the original data together with the estimated trend. (8 Marks)

(b) Using periods 1–12 only, calculate seasonal indices for both additive and multiplicative decomposition. (7 Marks)

(c) Produce forecasts for periods 13–16 using both models and compare their accuracy using one suitable error measure. (6 Marks)

(d) Using the better model, forecast periods 17–20 and justify your model choice briefly. (4 Marks)

Total 25 Marks

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2.

Hock-Brooke Automobiles receives new cars at the start of each week. Cars are stored in a warehouse and any excess above warehouse capacity must be placed in extra rented parking spaces.

The weekly number of cars delivered and the weekly customer demand are uncertain.

Cars Delivered	7	8	9	10
Probability	0.10	0.20	0.50	0.20

Weekly Demand	6	7	8	9	10	11
Probability	0.05	0.20	0.30	0.10	0.20	0.15

Each warehouse space costs £10 per week. Each extra rented parking space costs £20 per car per week.

Use the following random numbers for a 12-week simulation:

Week	1	2	3	4	5	6	7	8	9	10	11	12
Delivery RN	0.086	0.237	0.801	0.582	0.094	0.433	0.479	0.160	0.735	0.114	0.391	0.517
Demand RN	0.431	0.587	0.738	0.956	0.284	0.649	0.696	0.293	0.001	0.973	0.298	0.314

(a) Describe how you would model this problem as a simulation study. Your answer should identify the random inputs, the main state variable(s) and the performance measure(s). (5 Marks)

(b) Simulate 12 weeks of operation for a warehouse capacity of 9 cars. Your working should show the weekly deliveries, demand, stock at the start of the week, stock at the end of the week and total weekly cost. (8 Marks)

(c) Determine the average weekly cost for a capacity of 9 cars. (3 Marks)

(d) Repeat the simulation for capacities 7, 8 and 10 using the same random numbers. Present the average weekly costs in a clearly labelled table and identify the best capacity among 7, 8, 9 and 10. (6 Marks)

(e) State three reasons why a manager should be cautious about choosing a final warehouse size from just one short simulation run. (3 Marks)

Total 25 Marks

3.

F.B. Knight distributes goods from three warehouses to three shops. The aim is to minimise total transport cost while meeting all shop demand.

Warehouse	A	B	C
Available stock	100	100	100

Shop	X	Y	Z
Demand	70	80	150

	X	Y	Z
A	8	7	5
B	12	3	9
C	10	11	7

(a) Formulate a transportation model for the direct-shipping problem, clearly defining the decision variables, objective function and supply / demand constraints. (8 Marks)

(b) Solve the direct-shipping model and state one optimal distribution plan together with the minimum total transport cost. (6 Marks)

(c) Determine how the solution changes if warehouse B cannot deliver to shop Y and warehouse A must deliver at least 5 units to shop Z. State the revised optimal plan and total cost. (5 Marks)

(d) Two intermediary distribution points D_1 and D_2 are now available. If this option is used, all flow must go from a warehouse to a distribution point and then from a distribution point to a shop; no direct shipping is allowed.

	D_1	D_2
A	4	5
B	3	2
C	6	4

	X	Y	Z
D_1	4	5	2
D_2	2	4	4

Formulate the transhipment extension and determine whether it is better to use one or both intermediary points. Give the resulting minimum total cost and the non-zero flows. (6 Marks)

Total 25 Marks

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4.

A company manufactures two products, X and Y, with profits of £36k and £34k respectively.

Each unit of X uses 4 units of raw material A and 0.2 units of raw material B. Each unit of Y uses 3 units of raw material A and 0.3 units of raw material B.

Each week the company has at most 40 units of raw material A and 3 units of raw material B available.

(a) Formulate a linear programming model for this production problem, clearly defining the decision variables, objective function and constraints. (6 Marks)

(b) Solve the model to determine the optimal weekly production plan and the maximum weekly profit. (6 Marks)

(c) Identify the binding constraints and determine the unused quantity of each raw material at the optimum. (4 Marks)

(d) Extend the model so that for every unit of product X, exactly 3 units of product Y must be produced. Reformulate the model. (5 Marks)

(e) Solve the revised model and compare the revised profit with the original optimum. (4 Marks)

Total 25 Marks

End of Paper